

Simulated E-Banking System

Team #3

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Software Requirements Specification Document

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1. Introduction

1.1 Purpose

The purpose of this SRS is to provide a contract between the client and contractor for developing a simulated online banking platform. It describes what the system is going to do and what constraints it will operate under. The intended audience of the SRS is both the client and contractor.

1.2 Scope

The software product to be produced is the *Simulated E-Banking System* (SEBS). SEBS will include, but is not limited to, secure user access, account visibility, and common financial workflows in a correctness- and security-focused environment. SEBS shall be a simulation and rely entirely on test data and a mocked backend. SEBS shall provide users with a secure and convenient way to perform banking tasks. Applying SEBS in this way has the benefits of increasing user retention and satisfaction.

1.3 Definitions, acronyms, and abbreviations

- 1) *SEBS*: the Simulated E-Banking System
- 2) *RBAC*: Role-based Access Control
- 3) *Frontend*: The user interface (UI)
- 4) *Backend*: Server-side infrastructure handling business logic

- 5) *API*: Application programming interface
- 6) *Session token*: Unique, encrypted string acting as a "pass" to identify a user's session
- 7) *MFA*: Multi-factor authentication

1.4 References

Any references in the SRS will appear as superscripts.

- 1) Rescorla, Eric. "HTTP Over TLS." *Internet Engineering Task Force RFC 2818* (May 2000). Accessed 12 March 2026. <https://datatracker.ietf.org/doc/html/rfc2818>.

1.5 Overview

The remainder of the SRS contains more detailed requirements. The client is mostly concerned with Section 2, *Overall Description*, which contains natural-language descriptions of the required functionality. Developers are most concerned with Section 3, *Specific Requirements*, which translates natural-language requirements found in Section 2 into formal, testable requirements.

2. Overall description

2.1 Product perspective

SEBS is a standalone simulated banking platform. It is not replacing any existing banking system and does not integrate with any operating financial infrastructure. It is a self-contained application on the web that only simulates the experience of an online banking platform. All system behavior is controlled by those who developed it.

2.1.1 System interfaces

SEBS interface is with a mocked backend server that simulates real banking operations. The frontend will communicate with the backend via a well-defined software interface (*See Section 2.1.4, Software Interfaces*). No real external systems will be connected or associated with the SEBS interface.

2.1.2 Interfaces

The SEBS uses a web-based UI that is accessible through a web browser. The views of the UI depend on the user's role in the banking system.

2.1.3 Hardware interfaces

SEBS has no hardware interface requirements.

2.1.4 Software interfaces

SEBS does not require any specific external software products.

2.1.5 Communications interfaces

All communication between the frontend and backend shall use HTTPS¹. Any session tokens shall be transmitted securely and stored correctly on the client side.

2.1.6 Memory constraints

The system does not involve any memory constraints.

2.1.7 Operations

The system shall support multiple modes of operations. Normal operations involve users being able to log in, perform their banking needs, and log out when they are complete. Administrators shall access logs, reports, and tools to support customers as needed. Sessions shall time out when users are inactive for a certain period of time for safety and security.

2.1.8 Site adaptation requirements

SEBS does not need a site-specific adaptation. It shall be able to be deployed on any standard web server environment. Configuration files shall be used to manage any environment-specific settings including database connections and the durations of session timeouts.

2.2 Product functions

SEBS shall provide the following functions. Users will be able to securely login, log out, recover their passwords, and opt-in to MFA. They will be able to display their account balance, account types, and summaries for both checking and savings accounts. A full view of deposits, withdrawals, transfers, and payments shall be available to users. Users shall be able to make transfers between authorized accounts and have the support to do it immediately or scheduled. Users shall be able to update their profile, change credentials, and see a log of recent login activity. Administrators shall be able to access logs and reporting tools.

2.3 User characteristics

Customers shall be the general users of the SEBS. It is assumed that they have basic familiarity with e-banking and web browsers. They do not require more than basic technological knowledge to use the system. They interact with their account, transactions, transfers, and payment features.

Administrators are internal users responsible for providing customer service, policy configuration, and audit review. They are assumed to have more technical literacy and familiarity with web administrative tools. They have elevated access compared to the customer.

2.4 Constraints

The system shall be a simulation only and not be connected to any real financial networks or banking APIs. Any and all financial data is generated internally and is not real. The system shall enforce role-based access control (RBAC) to prevent lower level users from accessing features and data. The system will comply with general security best practices, such as encrypted communication and highly secure password storage.

2.5 Assumptions and dependencies

It is assumed that users will have access to a web browser and a stable internet connection. It is also assumed that the backend server and database are available and running so the system can function properly. The system depends on the mocked backend to simulate the financial operations. It is assumed that any data in SEBS is mocked and not real financial data.

2.6 Apportioning of requirements

Some features that may be added in the future to SEBS:

- Multi-factor Authentication (MFA)
- Exporting of transaction history into a CSV or PDF

3. Specific Requirements

3.1 External Interface Requirements

This section describes the interfaces through which SEBS interacts with users and supporting components. These include the user interface as well as any required hardware, software, and communication interfaces.

3.1.1 User Interface

- **UI-REQ-001:** The system shall provide a web-based user interface accessible through a standard web browser.
- **UI-REQ-002:** The interface shall include a login page for user authentication and a dashboard displaying account balances and summary information after successful login.
- **UI-REQ-003:** The interface shall provide pages that allow users to view transaction history and perform transfers between accounts.
- **UI-REQ-004:** Navigation elements shall allow users to access these functions from the main dashboard.

- **UI-REQ-005:** Administrative users shall have access to additional pages for managing test accounts and reviewing system activity, which shall only be accessible to users with administrative privileges.

3.1.2 Hardware Interfaces

- **HARDWARE-REQ-001:** The system shall operate on a standard personal computer used for development and demonstration.
- **HARDWARE-REQ-002:** The application shall be locally hosted and accessed through a web browser on the host machine. No specialized hardware interfaces are required.

3.1.3 Software Interfaces

- **SOFTWARE-REQ-001:** Users shall access the system through a standard web browser that supports HTML-based interfaces.

3.1.4 Communications Interfaces

- **COMMS-REQ-001:** The system shall communicate with users through standard HTTP requests and responses between the web browser and the locally hosted web server.
- **COMMS-REQ-002:** All communication shall occur within the local host environment.

3.2 Functional Requirements

3.2.1 Customer

Customers are the standard users of SEBS who access and manage their simulated bank accounts through the web interface. Customers are able to authenticate, view account information, review transaction history, and perform basic account operations provided by the system.

3.2.1.1 User Authentication and Login

- **LOGIN-REQ-001:** The system shall provide a login page where customers can enter a username and password.
- **LOGIN-REQ-002:** The system shall validate the provided credentials against stored user account information.
- **LOGIN-REQ-003:** The system shall grant access to the account when valid credentials are entered.
- **LOGIN-REQ-004:** The system shall deny access when authentication fails.
- **LOGIN-REQ-005:** The system shall display an error message when invalid credentials are submitted.

3.2.1.2 View Account Summary and Balances

- **ACCOUNT-REQ-001:** The system shall allow customers to view a dashboard displaying their accounts.

- **ACCOUNT-REQ-002:** The system shall display the current balance for each account associated with the customer.
- **ACCOUNT-REQ-003:** The system shall display summary information for each account, including the account name and number.
- **ACCOUNT-REQ-004:** The system shall allow customers to select an account to view additional account details.

3.2.1.3 View Transaction History

- **TRANSACTION-REQ-001:** The system shall allow customers to view the transaction history for a selected account.
- **TRANSACTION-REQ-002:** Each transaction record shall include the transaction date, type, and amount.
- **TRANSACTION-REQ-003:** The system shall display transactions in chronological order.

3.2.1.4 Transfer Funds Between Accounts

- **TRANSFER-REQ-001:** The system shall allow customers to initiate a transfer between their accounts.
- **TRANSFER-REQ-002:** The system shall allow customers to select a source account and a destination account for the transfer.
- **TRANSFER-REQ-003:** The system shall require the user to enter a transfer amount.
- **TRANSFER-REQ-004:** The system shall verify that the source account has sufficient funds before completing the transfer.
- **TRANSFER-REQ-005:** The system shall reject a transfer and display an error message if the source account has insufficient funds for the transfer.
- **TRANSFER-REQ-006:** The system shall update the balances of both accounts after a successful transfer.
- **TRANSFER-REQ-007:** The system shall record the transfer as a transaction in the transaction history of both accounts.

3.2.1.5 Simulated Deposits and Withdrawals

- **TRANSACTION-REQ-004:** The system shall allow customers to perform simulated deposit operations on a selected account.
- **TRANSACTION-REQ-005:** The system shall allow customers to perform simulated withdrawal operations on a selected account.
- **TRANSACTION-REQ-006:** The system shall require the user to enter a transaction amount for deposits and withdrawals.
- **TRANSACTION-REQ-007:** The system shall verify that sufficient funds are available before completing a withdrawal.
- **TRANSACTION-REQ-008:** The system shall update the account balance after a successful deposit or withdrawal.
- **TRANSACTION-REQ-009:** The system shall record deposits and withdrawals in the account transaction history.

3.2.2 Administrator

Administrators manage and monitor the operation of the system. Administrative users have access to functions for maintaining simulated users accounts and reviewing system activity.

3.2.2.1 Create and Manage Test Accounts

- **ADMIN-REQ-001:** The system shall allow administrators to create simulated customer accounts.
- **ADMIN-REQ-002:** The system shall allow administrators to modify account information for existing customers.
- **ADMIN-REQ-003:** The system shall restrict account management functions to users with administrator privileges.

3.2.2.2 Monitor System Activity

- **ADMIN-REQ-004:** The system shall allow administrators to view system activity related to user accounts.
- **ADMIN-REQ-005:** The system shall allow administrators to view records of simulated transactions performed within the system.
- **ADMIN-REQ-006:** The system shall restrict system monitoring functions to users with administrator privileges.

3.2.2.3 View Transaction Logs

- **ADMIN-REQ-007:** The system shall allow administrators to view transaction logs generated by customer account activity.
- **ADMIN-REQ-008:** The system shall display transaction records including the transaction date, type, and amount.
- **ADMIN-REQ-009:** The system shall restrict access to transaction logs to users with administrator privileges.

3.3 Performance Requirements

- **PERF-REQ-001:** The system shall respond to user requests within a reasonable time under normal operating conditions.
- **PERF-REQ-002:** The system shall update account balance and transaction records immediately after a completed operation.
- **PERF-REQ-003:** The system shall maintain acceptable performance when handling typical user actions at login, viewing account information, and performing transactions.

3.4 Logical Database Requirements

- **DB-REQ-001:** The system shall maintain a database for storing simulated customer accounts.
- **DB-REQ-002:** The system shall store account information including account identifiers and current balances.

- **DB-REQ-003:** The system shall maintain records of transactions associated with each account.
- **DB-REQ-004:** The system shall store user authentication information required for system login.

3.5 Design Constraints

- **DESIGN-REQ-001:** The system shall operate as a locally hosted web application.
- **DESIGN-REQ-002:** The system shall not connect to external banking systems or financial networks.
- **DESIGN-REQ-003:** The system shall use simulated data for all accounts and transactions.

3.5.1 Standards compliance

- **STD-REQ-001:** The system documentation shall follow the IEEE standards used for project documentation.
- **STD-REQ-002:** The Software Project Management Plan shall follow IEEE Std 1058.
- **STD-REQ-003:** The System Requirements Specification shall follow IEEE recommended structure for software requirements documentation.

3.6 Software System Attributes

This section describes the non-functional attributes of the system. These attributes define qualities such as reliability, availability, security, maintainability, and portability that SEBS must satisfy.

3.6.1 Reliability

- **REL-REQ-001:** The system shall maintain consistent account balances and transaction records during normal system operation.
- **REL-REQ-002:** The system shall prevent incomplete or failed transactions from modifying stored account data.

3.6.2 Availability

- **AVAIL-REQ-001:** The system shall be available for use whenever the locally hosted application server is running.
- **AVAIL-REQ-002:** The system shall allow users to access the application through a web browser while the system is active.

3.6.3 Security

- **SEC-REQ-001:** The system shall require users to authenticate with a valid username and password before accessing account functions.

- **SEC-REQ-002:** The system shall restrict administrative functions to users with administrator privileges.
- **SEC-REQ-003:** The system shall protect stored user credentials using secure storage methods.

3.6.4 Maintainability

- **MAINT-REQ-001:** The system shall be structured in modular components to support future updates and modifications.
- **MAINT-REQ-002:** The system source code shall be maintained in a version control repository.
- **MAINT-REQ-003:** The system documentation shall be updated to reflect changes made to system functionality.

4. Supporting information

- N/A